

MAL-Analytics

Automated Web Scraping for Anime Intelligence

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Project Overview

This project leverages Python to bridge the gap between static web content and structured data analytics.

By targeting **MyAnimeList**, we've built a robust extraction engine that harvests real-time rankings, user scores, and series metadata for the top 200 titles globally.

The goal is to transform messy HTML into a clean, research-ready dataset for trend analysis and recommendation modeling.



The Tech Stack



Requests

Handles the HTTP protocol to "fetch" raw HTML from the MAL servers with custom headers.



BeautifulSoup

Parses the raw HTML document into a searchable tree for precise data pinpointing.



Pandas

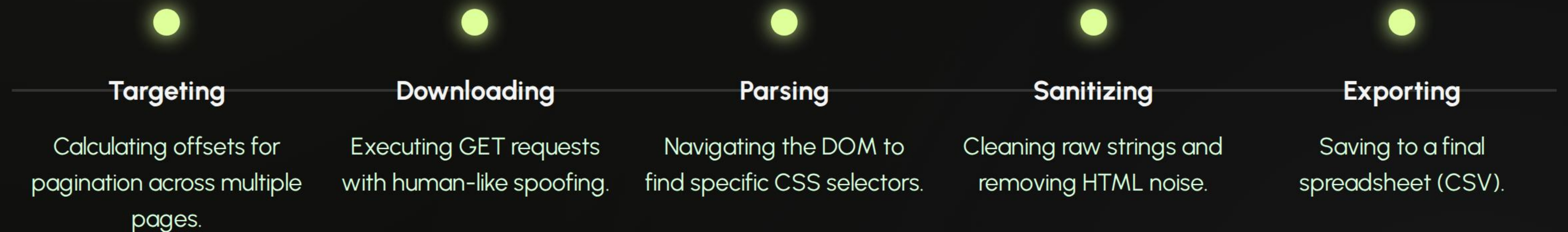
Acts as the "Accountant," converting raw lists into a structured CSV database format.



Time

Manages rate-limiting to ensure the script behaves ethically and prevents IP bans.

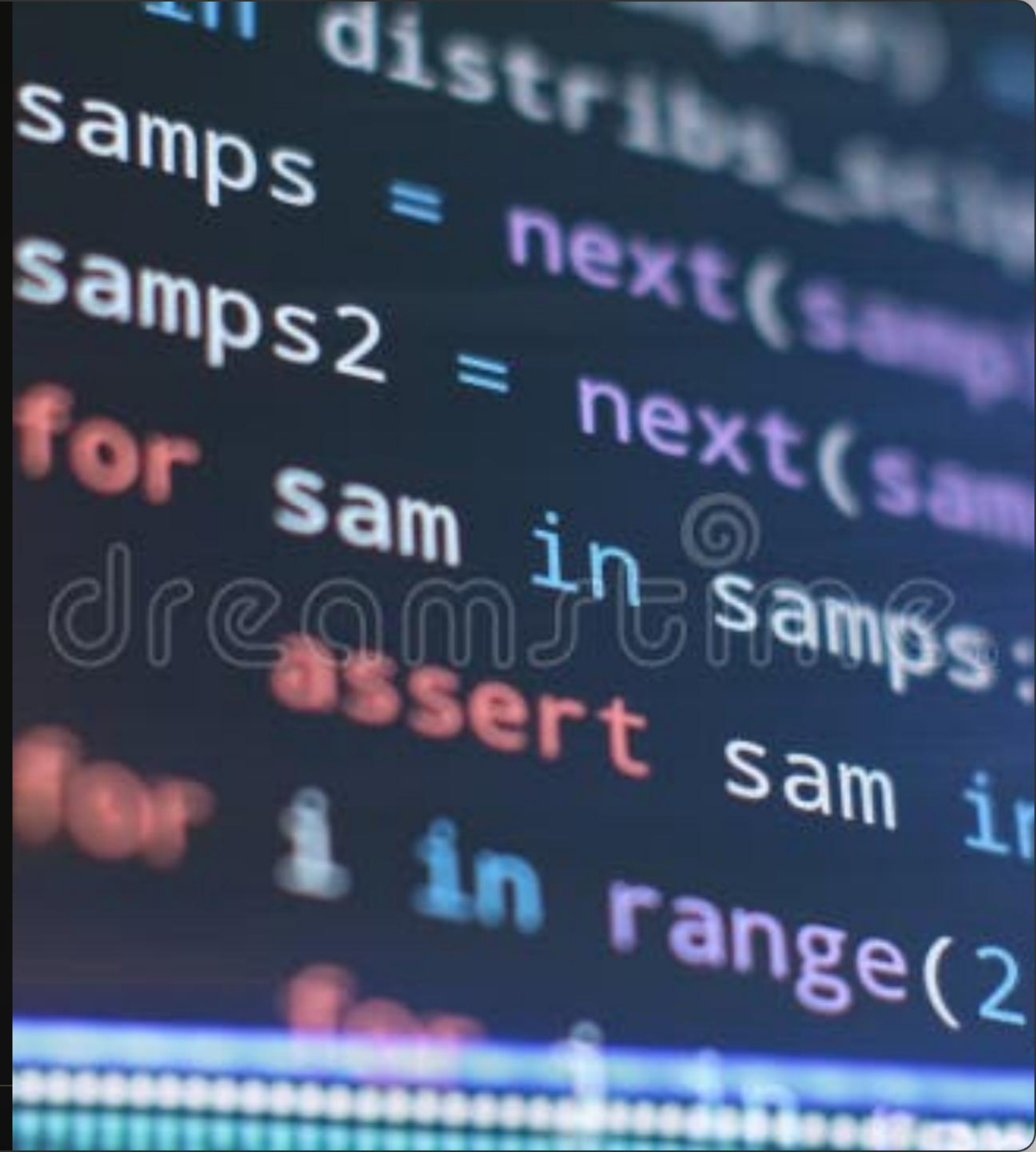
The Pipeline



Mimicking Human Behavior

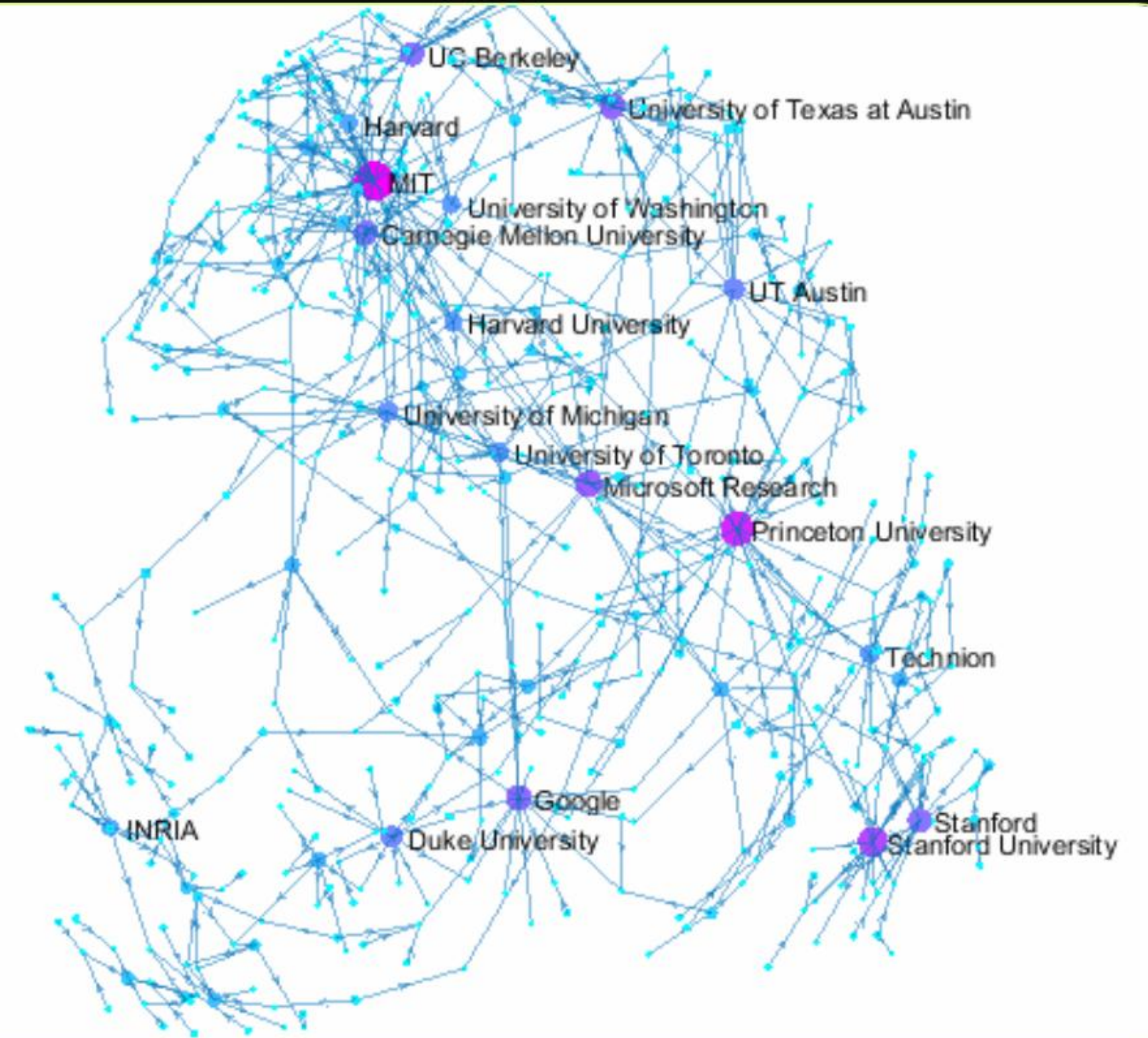
Websites often block basic bots. We bypass this using **Header Spoofing** to simulate a real browser interaction.

The script uses a dynamic **Offset Loop** to visit pages in chunks of 50, allowing us to seamlessly scale the data collection from 50 to 200+ records without manual intervention.



Navigating DOM Architecture

- Q Row Isolation:** Targeting every tr tag with the ranking-list class.
- >_ Tag Drilling:** Locating titles within h3 tags and ratings in specific span elements.
- 🔑 Metadata Extraction:** Digging into the nested "information" div to split air dates and episode counts.



Data Sanitization

Cleaning Methods

.strip(): Removes invisible newlines and spaces that break CSV alignment.

.split(): Separates "Media Type" from "Episodes" in shared HTML containers.

.replace(): Strips away labels like "members" and commas for numeric analysis.

✓ Before vs After

Raw: "\n TV (64 eps) "

Cleaned: "TV" | "64"

Raw: "3,210,123 members"

Cleaned: 3210123

System Robustness

-  **Fault Tolerance:** Nested try-except blocks ensure that if one anime has missing data, the script skips that row rather than crashing the entire process.
-  **Network Resiliency:** 10-second timeouts prevent the program from hanging on slow connections.
-  **Graceful Shutdown:** The finally block ensures data is saved to CSV even if the user manually stops the script (KeyboardInterrupt).

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To remain a "Good Web Citizen," we implement **Rate Limiting**. By pausing for 2 seconds between every page request, we avoid triggering server alerts and ensure we don't disrupt the site's performance for other users.

1	Fullmetal Alchemist: B	TV	64	9.10	3200000
2	Steins;Gate	TV	24	9.07	2500000

Real World Impact

This technology is used daily by streaming giants to track competitor popularity, identify trending genres, and build intelligent recommendation

Thank you for your time!

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Thank you for your time!

Image Sources



https://miro.medium.com/v2/resize:fit:1400/0*tatt3yEupliGl6Gz.jpeg

Source: medium.com



Thumbnail for <https://thumbs.dreamstime.com/b/data-science-machine-learning-programming-code-screen-abstract-technology-background-giant-blurred-python-script-456387070.jpg>

Source: es.dreamstime.com



<https://blogs.mathworks.com/images/loren/2017/nips2015.gif>

Source: blogs.mathworks.com
